

1. Proof of concept of XPCS for grain-scale dynamics in SSEs
2. B10 vs. mixed-anion: local strain from anion distribution causes fluctuation in sub-domains
 - Clustering effects of local B12-rich environment might have significantly different lattice spacing -> strain
3. Checkerboard patterning = sub-domain wall boundary fluctuation (i.e., analogy to crystal growth)
 - Heterodyning from surface vs. bulk scattering ? Is this the same physical effect as domain exchange?
4. dQ dependence -> more defective or strained regions exhibited stronger checkerboard dynamics
5. [maybe] Temperature effects -> glass transitions in Na sublattice vs. anion rotation vs. Na conduction
 - Quantitative analysis of the two-time correlation plots – able to derive a quasi-velocity / frequency?
 - How does the velocity / frequency change with temperature? Activation barrier? Etc.
 - Regression of the Langevin to quantitatively describe the two-time correlations
 - Analysis over well-defined transitions – which part of the physics are changing the most?
 - ~120K vs. 300K (from memory)

Figure 1. Sample characteristics / data acquisition geometry

Cartoon of B10 low temp

Cartoon of B10 high temp

XRD data for phase confirmation

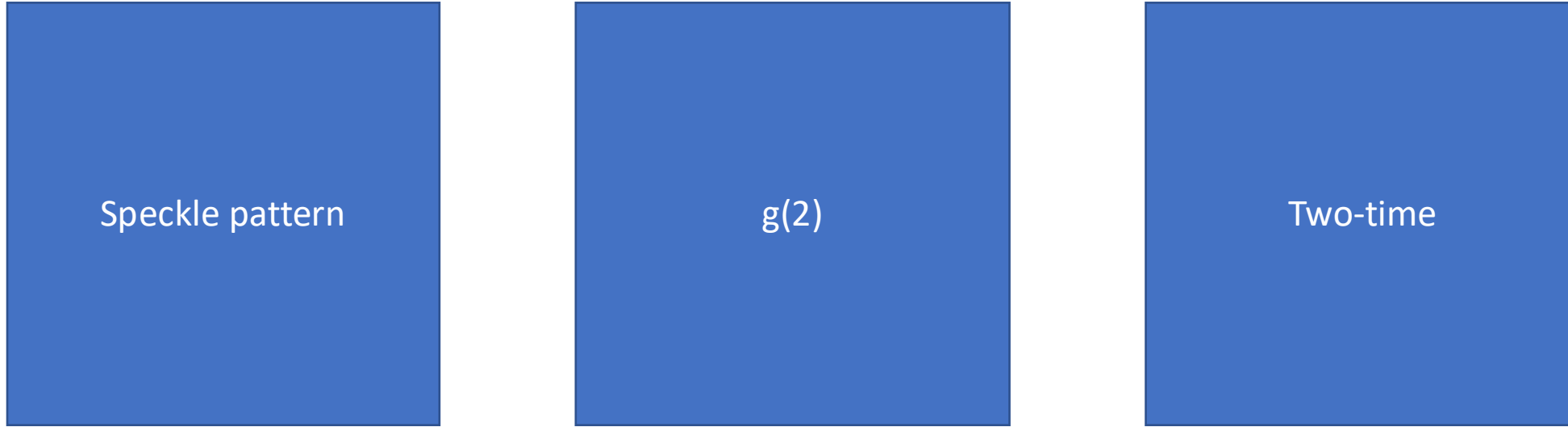
Data acquisition

Cartoon of mixed low temp (local strain sub-domains are shown explicitly)

Cartoon of mixed high temp (Na sublattice delocalization with transport)

Temperature-dependent conductivity

Figure 2. Data processing workflow & interpretation



How the $g(2)$ is graphically related to two-time

Figure X. Q (and phi?) dependence of dynamics (room T)



Figure X. Temperature-dependence of two-times (selected ROI)

